

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2236

Hutchins 138 kV
January 22, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2236. The Interconnection Customer proposes a 177.9 MW solar photovoltaic facility interconnecting at the Hutchins 138 kV substation in the Virginia area. The request was received on November 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 417 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,492,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Decatur to Hutchins 138 kV	Clover 345/138 kV Auto	112.7	213	35.9
Clover 345 kV/138 kV Auto	Bauxite to Hutchins 138 kV	110.2	673	79.8
Layfair to N Decatur 138 kV L	Clover 345/138 kV TX	108.5	212	35.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Decatur to Hutchins 138 kV	Rebuild line with higher-capacity conductor	\$24,100,000	23% (\$5,613,000)
Clover 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$12,491,000	22% (\$2,805,000)
Layfair to North Decatur 138 kV	Reconductor line	\$8,905,000	23% (\$2,074,000)
	Total allocated to project		\$10,492,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.